

Quantifying SZZ-Induced Label Noise in Just-In-Time Software Defect Prediction under Verification-Latency-Aware Online Evaluation

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Overview

- Just-in-time software defect prediction (JIT-SDP) predicts, at commit time, whether a code change is likely to introduce a defect, so that review and testing effort can be directed to risky changes [6].
- The models are trained on past commits. Each commit is described by change metrics and is labelled as clean or defect-introducing.
- These labels are not recorded in the repository. They are inferred by the SZZ algorithm, which traces the lines changed in a bug-fixing commit back to the commits that last modified them [10].
- Bug-fixing commits often contain unrelated changes (tangled commits), so SZZ produces wrong labels. Only 17% to 32% of the changed lines in bug-fixing commits contribute to the fix [5].

Overview

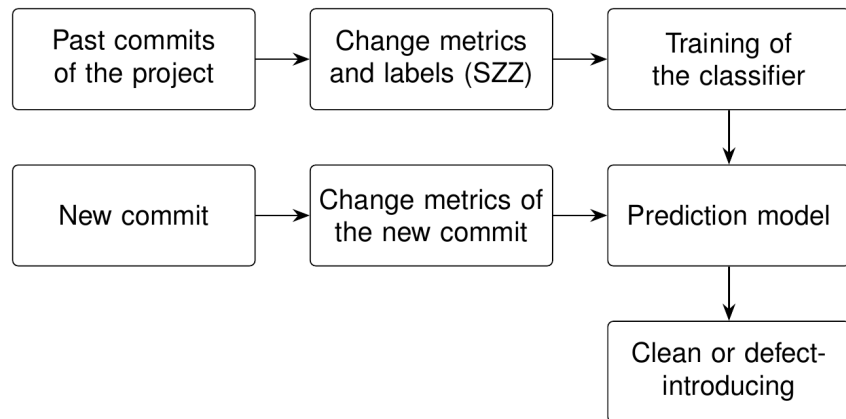


Figure 1: Workflow of JIT-SDP

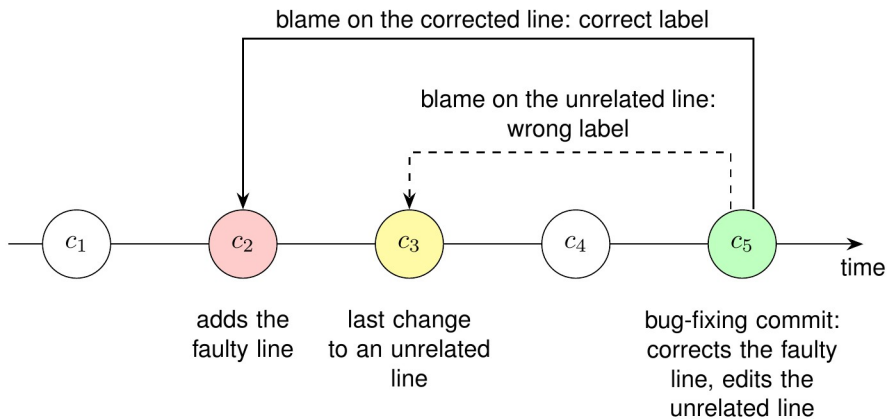


Figure 2: Labelling with SZZ and a wrong label caused by an unrelated change

Motivation

- JIT-SDP models are trained and tested on the same SZZ labels. A model is then rewarded for reproducing the errors of SZZ, and the effect of label noise on the reported performance is not visible.
- Random k-fold cross-validation is still widely used, although it trains the model on commits that are newer than the test commits [3].
- In practice the label of a commit arrives late (verification latency): a defect is recognised only when it is fixed, which can take months or years [1].
- It is therefore not known how much SZZ label noise costs a model that is evaluated in a realistic setting.

Related Works - 1

Title	Author	Dataset	Key Takeaways
When Do Changes Induce Fixes? (2005) [10]	Jacek Śliwerski	Mozilla and Eclipse	Proposes the SZZ algorithm: lines changed by a bug fix are traced back with annotate (blame) to find the bug-introducing changes
A Framework for Evaluating the Results of the SZZ Approach for Identifying Bug-Introducing Changes (2017) [2]	Daniel A. da Costa	Ten open-source projects	Criteria for evaluating SZZ implementations; proposes MA-SZZ, which ignores meta-changes
Evaluating SZZ Implementations Through a Developer-Informed Oracle (2021) [9]	Giovanni Rosa	1,930 bug-fixing commits in which the developer names the introducing commit	Compares SZZ variants against the oracle; R-SZZ performs best (precision about 66%); releases PySZZ
A Fine-grained Data Set and Analysis of Tangling in Bug Fixing Commits (2022) [5]	Steffen Herbold	LLTC4J: manually labelled bug-fixing commits of Apache Java projects	Only 17% to 32% of the changed lines contribute to the bug fix; tangling adds noise to defect data
The Impact of Changes Mislabeled by SZZ on Just-in-Time Defect Prediction (2021) [4]	Yuanrui Fan	126,526 changes from ten Apache projects	Compares four SZZ variants with RA-SZZ as the reference; labels of AG-SZZ reduce performance significantly

Related Works - 2

Title	Author	Dataset	Key Takeaways
A Large-Scale Empirical Study of Just-in-Time Quality Assurance (2013) [6]	Yasutaka Kamei	Six open-source and five commercial projects	14 change metrics in five groups; 68% accuracy and 64% recall with logistic regression
Are Fix-Inducing Changes a Moving Target? (2018) [7]	Shane McIntosh	Qt and OpenStack (37,524 changes)	Properties of fix-inducing changes change over time; models lose predictive power as they age
Class Imbalance Evolution and Verification Latency in Just-in-Time Software Defect Prediction (2019) [1]	George G. Cabral	Ten GitHub projects	Online JIT-SDP with delayed labels (90-day waiting period); proposes ORB for evolving class imbalance
Deep Just-in-Time Defect Prediction: How Far Are We? (2021) [11]	Zhengran Zeng	Open-source projects with over 310,000 changes	Deep models do not consistently beat simple ones; LAPredict (logistic regression on added lines) is best in most cases
The Best of Both Worlds: Integrating Semantic Features with Expert Features for Defect Prediction and Localization (2022) [8]	Chao Ni	JIT-Defects4J: 21 Java projects, 27,319 commits	Dataset built from manually validated bug-fixing lines; proposes JIT-Fine

Research Gaps

- Noise of SZZ labels not measured over complete project histories
- Models tested against the same labels on which they are trained
- Effect of label noise not separated from the effect of the evaluation procedure
- Label noise not studied under verification latency
- Effect of SZZ noise on online learners not understood, and no remedy for this setting

Research Questions

- **RQ1:** How much do the labels of SZZ variants disagree with one another and with independent reference labels, and what type of error does each variant make?
- **RQ2:** How does the choice of the label source affect the measured performance of JIT-SDP models under different evaluation settings?
- **RQ3:** Through which mechanism does label noise reduce the performance of an online learner, and which type of error is responsible?
- **RQ4:** Can an online learner be made less sensitive to SZZ label noise without access to clean labels?

Objectives

- **Objective 1:** Generate the labels of six SZZ variants for a dataset with independent reference labels, and characterise their false positive and false negative rates.
- **Objective 2:** Examine the agreement between the variants and identify the sources of their errors.
- **Objective 3:** Measure how much the evaluation procedure changes the measured performance of JIT-SDP models.
- **Objective 4:** Measure the loss caused by SZZ labels for an online learner under verification latency.
- **Objective 5:** Identify the type of labelling error responsible for the loss and its mechanism.
- **Objective 6:** Design and evaluate a noise-aware online learner.

Dataset

- JIT-Defects4J [8]: 27,319 commits from 21 Apache Java projects (2001 to 2019)
- 2,332 commits (8.54%) are defect-introducing;
5,453 commits are bug-fixing
- Reference labels: bug-fixing lines validated by at least three of four annotators, then traced back with git blame
- Features: the 14 change metrics of Kamei et al. [6]
- Selected because its labels do not come from the SZZ variants under study
- SZZ labels and label arrival times will be added in MTP II

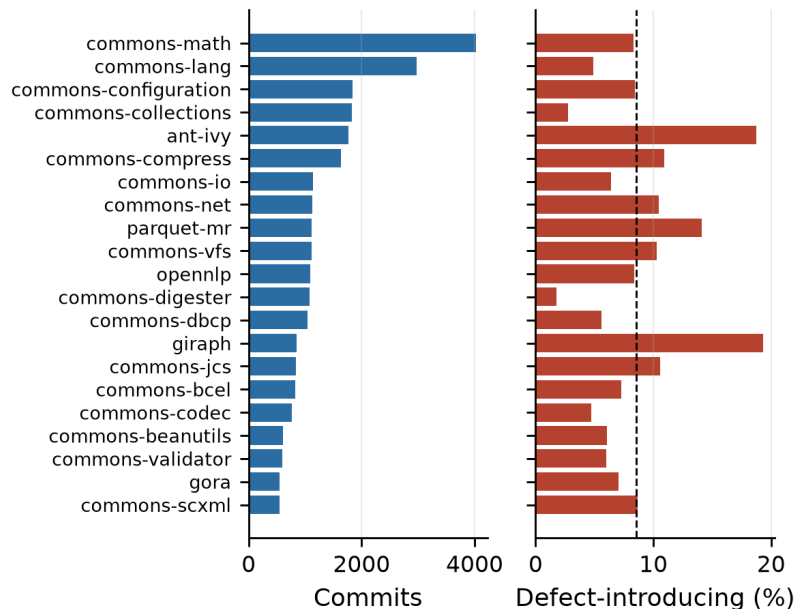


Figure 3: Commits and defect rate of each project

Proposed Work

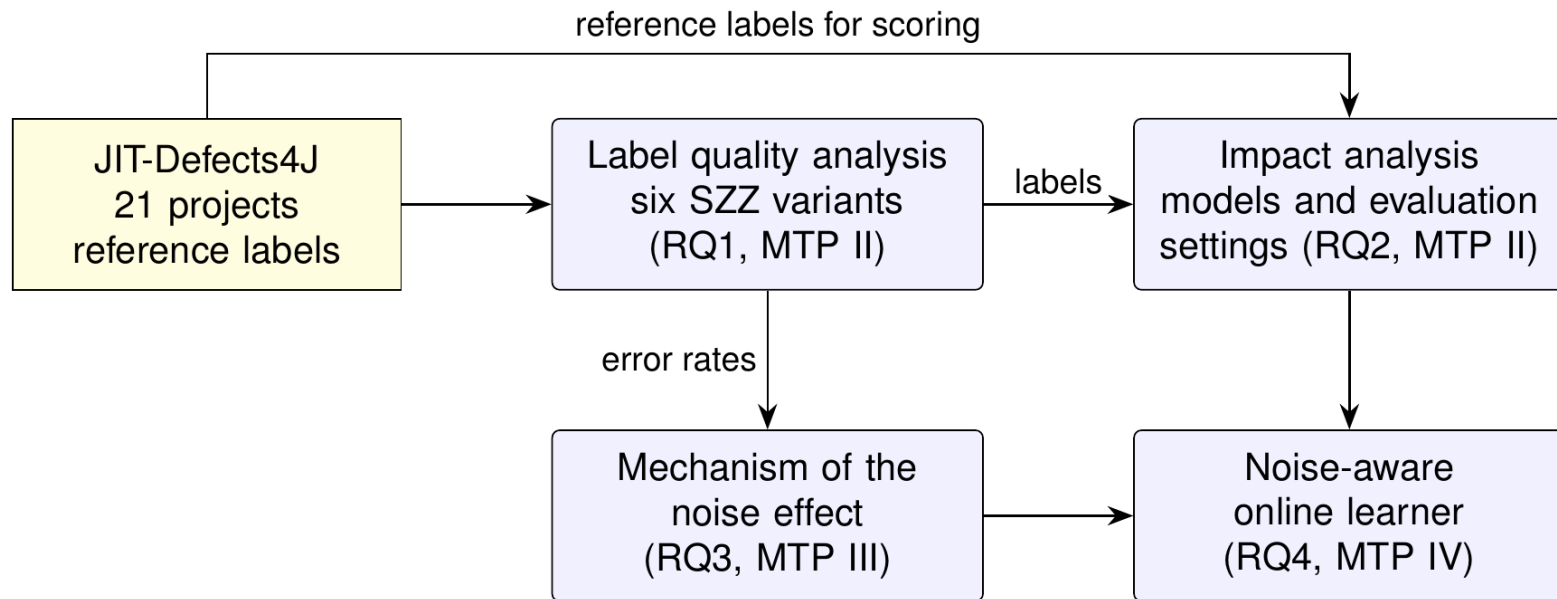


Figure 4: Planned studies and the research questions they address

Timeline

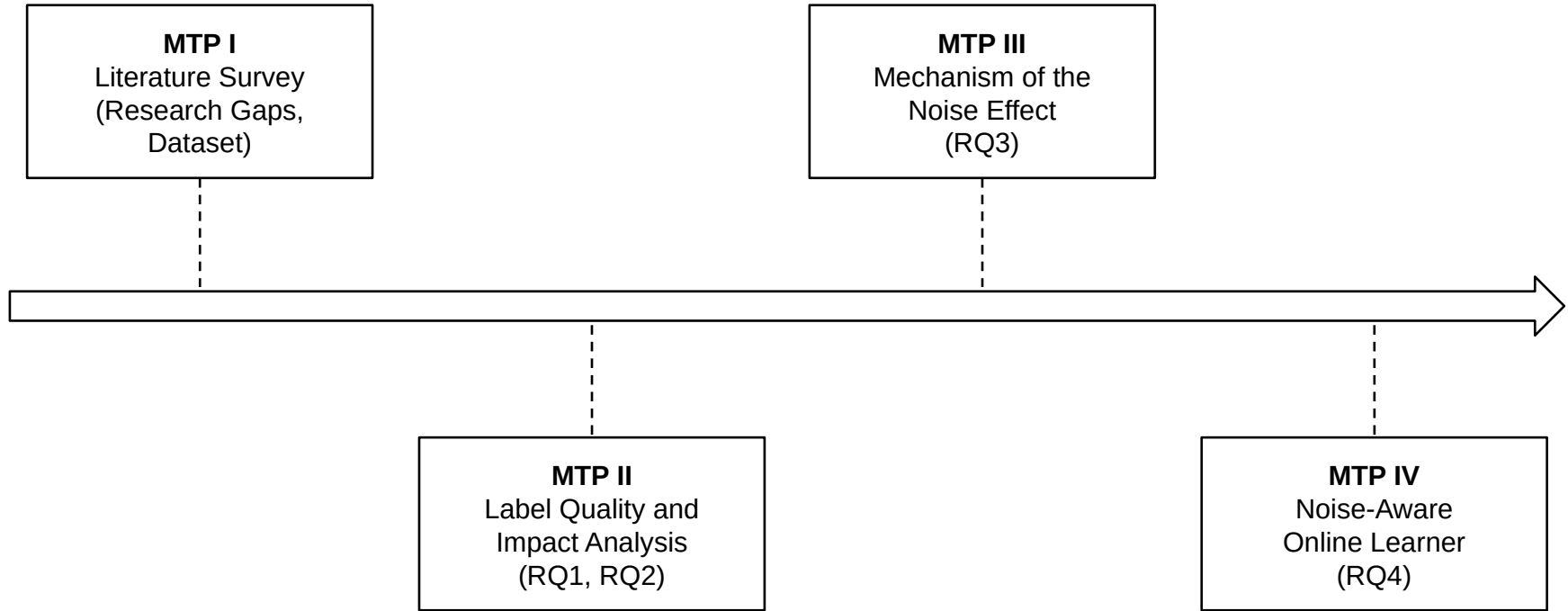


Figure 5: Timeline

References I

- [1] George G. Cabral et al. “Class Imbalance Evolution and Verification Latency in Just-in-Time Software Defect Prediction”. In: *Proceedings of the 41st International Conference on Software Engineering (ICSE)*. 2019, pp. 666-676.
- [2] Daniel Alencar da Costa et al. “A Framework for Evaluating the Results of the SZZ Approach for Identifying Bug-Introducing Changes”. In: *IEEE Transactions on Software Engineering* 43.7 (2017), pp. 641-657.
- [3] Davide Falessi et al. “On the Need of Preserving Order of Data When Validating Within-Project Defect Classifiers”. In: *Empirical Software Engineering* 25 (2020), pp. 4805-4830.
- [4] Yuanrui Fan et al. “The Impact of Changes Misabeled by SZZ on Just-in-Time Defect Prediction”. In: *IEEE Transactions on Software Engineering* 47.8 (2021), pp. 1559-1586.
- [5] Steffen Herbold et al. “A Fine-grained Data Set and Analysis of Tangling in Bug Fixing Commits”. In: *Empirical Software Engineering* 27.6 (2022), p. 125.
- [6] Yasutaka Kamei et al. “A Large-Scale Empirical Study of Just-in-Time Quality Assurance”. In: *IEEE Transactions on Software Engineering* 39.6 (2013), pp. 757-773.
- [7] Shane McIntosh and Yasutaka Kamei. “Are Fix-Inducing Changes a Moving Target? A Longitudinal Case Study of Just-in-Time Defect Prediction”. In: *IEEE Transactions on Software Engineering* 44.5 (2018), pp. 412-428.
- [8] Chao Ni et al. “The Best of Both Worlds: Integrating Semantic Features with Expert Features for Defect Prediction and Localization”. In: *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*. 2022, pp. 672-683.

References II

- [9] Giovanni Rosa et al. “Evaluating SZZ Implementations Through a Developer-Informed Oracle”. In: *Proceedings of the 43rd International Conference on Software Engineering (ICSE)*. 2021.
- [10] Jacek Śliwerski, Thomas Zimmermann, and Andreas Zeller. “When Do Changes Induce Fixes?”. In: *Proceedings of the International Workshop on Mining Software Repositories (MSR)*. 2005, pp. 1-5.
- [11] Zhengran Zeng et al. “Deep Just-in-Time Defect Prediction: How Far Are We?”. In: *Proceedings of the 30th ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*. 2021.